

PRESENTATION SCRIPT

Modelling Phase Results

Worth the Investment? Weekly Stakeholder Update

Two Angles on Player Value and Injury Risk

Thursday, September 17, 2026 · Presented by Mahmoud
Script follows the "SH3_Modelling_Phase_Results.pptx" deck,
slide by slide

How to use this script

- Each card below matches one slide, in order. The gold number is the slide number, so it is easy to check you are on the right one.
- The main paragraph is what to say out loud. It is written in plain, spoken language, not slide text, so it will not sound like you are reading bullet points.
- The gold "if asked" boxes are optional. Only use them if a stakeholder actually asks a follow up question on that slide.

- You do not need to say every line word for word. Treat this as a safety net, not a script to recite.
- This script uses "I" throughout, matching the deck, since this update is framed as your individual analysis work rather than a team narration.
- Slide 17, the code appendix, sits after the thank-you slide. It is not part of the main flow, only jump to it during questions if someone asks to see the actual code.
- The deck now has a kickoff-whistle sound cue on slide 1 and a final-whistle cue on slide 16. Both are noted in the PowerPoint's own speaker notes, so check those before presenting; the same "final whistle, then open for questions" note also appears on slides 5 and 8, which looks like a leftover from copying the closing slide and is worth a quick check before you present.

17

slides

~12

min talk time

9:00

prep, per usual
schedule

10:00

discussion
starts

What this meeting needs to cover

The instructor's Discord message for this Thursday asks for the modelling phase results, with at least two points of view, questions, or angles on the topic. More are welcome, but two is the floor. No specific tool is required.

This deck answers that directly: two full angles, one from Recruitment and Scouting, one from Sporting Directors, each with its own method, chart, and honest interpretation, plus a bonus third angle checking the value forecast model itself.

OPENING LINE, BEFORE SLIDE 1

"Good morning, thank you for the time. Today I'm sharing modelling results from two angles on my topic: whether injury history predicts a value drop, for recruitment and scouting, and whether age predicts a better value-to-risk tradeoff, for sporting directors. I also checked how well my value forecast model actually performs, as a bonus third angle. This will take about twelve minutes, and I'm happy to take questions any time."

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TITLE SLIDE

Modelling Phase Results

SAY

"This is my title slide. Still the Worth the Investment project, player value and injury risk, and this is my weekly stakeholder update for September 17th: modelling phase results, two angles on player value and injury risk."

SAY

"Quick roadmap check before the numbers. Raw analysis, the value and injury patterns from last week, is done. This week is modelling and hypothesis testing: putting two of my stakeholder questions through real statistical tests. Dashboards and the final presentation are up next."

IF ASKED HOW THIS DIFFERS FROM LAST WEEK'S RAW ANALYSIS

Last week showed patterns: value tends to peak in a certain age range, injuries are unevenly distributed. This week tests specific stakeholder questions against those patterns with real statistics, so the difference is descriptive versus tested.

SAY

"Two required angles today, headlines first, evidence in a moment. For recruitment and scouting: does a player's injury history predict a value drop? Yes, but any injury counts, not just recent ones. For sporting directors: does age predict a better value-to-risk tradeoff? Risk, yes; value growth only in the 21 to 24 window. I'll also share a bonus third angle: my own value forecast is right just 22.7 percent of the time against a 50 percent baseline, so it needs rebuilding."

A Quick Note on the Statistics

SAY

"Quick note before the results, since both tests are new this week. Angle 1 uses a Mann-Whitney U test: it ranks every value instead of averaging, so outliers can't skew the result. Angle 2 uses a Kruskal-Wallis test: same idea, extended past two groups, since I'm testing five age buckets at once. Both ask one thing: is the difference real, or could chance produce it? With samples this large, p under 0.0001 is easy to clear, so it tells me a difference exists, not how big it is."

IF ASKED WHY NOT A SIMPLER TEST, LIKE A T-TEST OR ANOVA

A t-test and ANOVA assume roughly normal, bell-curve data. This data is skewed, a few huge outliers and a lot of near-zero values, so a mean-based test would be misled. Mann-Whitney and Kruskal-Wallis rank the data instead, the honest choice for this shape.

IF ASKED FOR THE EFFECT SIZE, NOT JUST THE P-VALUE

Fair catch, and I don't have it yet. Both notebooks report the test statistic and the p -value only, no rank-biserial correlation or epsilon-squared. That's the natural next number to add before this goes into the final report.

5

ANGLE 1, SECTION MARKER

Angle 1: Recruitment and Scouting

SAY

"First angle: recruitment and scouting. Does a player's injury history predict a value drop?"

Does Recent Injury History Predict a Value Drop?

SAY

"Which rising-value targets carry hidden injury risk? For every valuation snapshot, I looked at whether the player had a prior injury, and if so, how recently, then compared the value change at that snapshot. I bucketed by recency: within 90 days, 91 to 365 days, over 365 days, or no prior injury at all. Then I ran a Mann-Whitney U test comparing recent-injury snapshots against all others, since value change is skewed, not normally distributed, so ranks beat averages here."

IF ASKED WHY MANN-WHITNEY INSTEAD OF A T-TEST

A t-test assumes roughly normal, symmetric data. Value change is heavily skewed, a few huge outliers and a lot of near-zero changes, so Mann-Whitney, which compares ranks rather than means, is the honest choice here.

Any Injury History, Not Its Recency, Separates the Groups

SAY

"Here's the result. Three injured buckets, one no-injury baseline. Every injured bucket, recent or years old, sits at a flat **0.00 percent** median value change. Only the no-injury baseline shows real growth, **11.11 percent**, p less than 0.0001. This surprised me, I expected a gradient. Instead, timing barely matters, it's any injury history at all that separates the groups. Excluding likely breakout-debut snapshots, the no-injury median softens to **4.35 percent**, but the pattern holds."

IF ASKED WHETHER THIS PROVES INJURIES CAUSE LOWER VALUE GROWTH

No. This is a correlation from real data, not a controlled experiment, injury history could be standing in for playing style, age, or role. I'd treat it as a strong signal worth modelling, not proof of causation.

IF ASKED FOR THE EXACT, UN-ROUNDED 0.00 PERCENT FIGURE

That's the rounded display value. I haven't pulled the exact number yet, it's on my list alongside the effect size.

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ANGLE 2, SECTION MARKER

Angle 2: Sporting Directors

SAY

"Second angle: sporting directors. Does age predict a better value-to-risk tradeoff?"

Does Age Predict a Better Value-Risk Tradeoff?

SAY

"Are younger signings really the lower-risk, higher-upside bet? For each player, I paired their median value growth with their injury rate per 1,000 90s played, then grouped players into five age bands. I excluded 1,310 players with zero recorded playing time, plus 67 more with under 10 total 90s played, since a rate needs a real denominator, that leaves 582 players. I ran a Kruskal-Wallis test on both value growth and risk rate, the right test for comparing more than two skewed groups."

IF ASKED WHY NOT JUST TREAT THE EXCLUDED PLAYERS AS ZERO RISK

Because I don't actually know their risk, I only know I don't have playing-time data for them. Setting it to zero would silently claim they're the safest players in the dataset, which isn't something the data supports. Excluding and disclosing is the more honest choice.

Age Predicts Risk. It Doesn't Predict Value Growth.

SAY

"Two charts, same four usable age buckets, Under-21 is excluded, only 4 players. On the left, median value growth: only the 21 to 24 emerging bracket shows real positive growth, **7.14 percent**, everything else sits at zero. On the right, injury risk more than doubles, from **2.49** per 1,000 90s in the 21 to 24 bracket up to **5.95** in the 33-plus bracket, though it isn't perfectly ordered, 25 to 28 sits slightly above 29 to 32. Both patterns are statistically significant, but with a sample this large that mainly confirms the differences are real, not how large they are, effect sizes are still on my list to report."

IF ASKED WHY TWO SEPARATE CHARTS INSTEAD OF ONE

Value growth is a percentage and injury risk is a rate per 1,000 90s, different units on very different scales, so overlaying them on one axis would distort both. Keeping them side by side lets each speak for itself.

Bonus: The Forecast Model Is Reliably Wrong

SAY

"A third angle, briefly: model validation. I already have a Prophet forecast built for player value, so I stress-tested how well it actually performs before I'd let myself lean on it for anything. The honest number: **22.7 percent** directional accuracy against a 50 percent coin-flip baseline, and a **149.4 percent** median magnitude error, the forecast misses by more than the true value itself. It's also biased toward optimism, 91 percent of predictions say rising, but 80 percent of the actual holdout snapshots were falling. The fix is straightforward: age and injury history, from today's two angles, need to become real input features before this model earns any trust."

IF ASKED WHY I'M SHOWING A MODEL THAT DOESN'T WORK

Because a negative result is still a finding. It points straight at the fix rather than hiding a weak model, and it means I won't be leaning on this forecast for anything until age and injury history are built into it.

SAY

"Pulling this together: history matters more than timing, any past injury goes with flatter value growth, not just a recent one, though I still want to confirm how much of that flatness is valuations that simply never moved. And age is a risk signal, not a value rule, injury risk is higher for older players, though not strictly in order, and value growth follows no usable age rule on its own. Together, that's a recruitment signal built on injury history and age as separate inputs, not on one feature standing in for both."

SAY

"Three limits I'm carrying forward. Small samples in places, the Under-21 bucket is only 4 players, and the full age-risk comparison drops to 582 players once I require real playing time. The playing-time gap is still here, **1,310 of 1,959 players, 67 percent**, have no matching on-field performance record, the same coverage gap from last week. And the flat medians on Angle 1 aren't fully verified yet, every injured bucket sits at exactly 0.00 percent, which points to snapshots where the value simply never moved, and I still need to report what share of snapshots actually change at all."

IF ASKED WHETHER 0.00 PERCENT MEANS LITERALLY NO CHANGE

Not necessarily. It's the rounded median across each bucket, and I haven't yet broken out what fraction of snapshots in those buckets show any change at all versus a valuation that was simply never updated. That breakdown is the honest next step before I'd treat 0.00 percent as a precise finding.

SAY

"Before I close, a quick preview of where this goes next. On the left is a live Tableau dashboard, the same Recency and Position Risk Profile view behind today's first angle. Production, output visualisations, and the story concept for the final presentation each get their own session next week, so this is deliberately just a taste, not the full picture."

IF ASKED TO SEE MORE OF THE DASHBOARDS NOW

I'd rather give that its own proper session next week than rush it here, this slide is just a preview so today stays focused on the two tested angles.

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PHASE RECAP

Modelling and Hypothesis Testing: 100% Complete

SAY

"Quick recap before wrapping up. This phase is complete: I put both stakeholder questions through rank-based statistical tests, found that any injury history, not its recency, tracks with flat value growth, showed injury risk more than doubling from the 21 to 24 bracket to 33-plus, stress-tested my own forecast and documented exactly where it fails, and flagged the flat-median and playing-time caveats in the open rather than glossing over them. That's phase 4 of 6, dashboards and presentation are up next."

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CLOSING

Next Update & Thank You

SAY

"My next update brings today's two angles into the Tableau dashboards and the final capstone deck. Thank you for your time, I'm happy to take questions now."

WRAPPING UP

MOVING INTO DISCUSSION

After slide 16, pause, take a breath, and let the room lead. It's fine to say "let me check my notes" before answering something specific, that reads as careful, not unprepared.

If a question reaches ahead into dashboards or the final presentation, it's fine to say plainly that those are the next update, today was about testing two of my stakeholder questions with real statistics.

If someone asks to see the actual code behind either test, that's slide 17, past this closing slide. It's a deliberate appendix, not something you present unprompted.

APPENDIX · ONLY IF ASKED

The Code Behind Both Tests

SAY

"Since it's been asked, here's the real code, not simplified for the slide. Mann-Whitney is one line once the two groups are split out. Kruskal-Wallis takes a list of groups, one per age bucket, unpacked with a star."

HYPOTHESIS_INJURY_RECENCY_VS_VALUE.IPYNB · ANGLE 1

```
recent = merged.loc[merged.recency_bucket == "recent  
(<=90 days)", "value_change_pct"]  
other  = merged.loc[merged.recency_bucket != "recent  
(<=90 days)", "value_change_pct"]  
u_stat, p_value = stats.mannwhitneyu(recent, other,  
alternative="less")
```

HYPOTHESIS_AGE_VS_VALUE_RISK_TRADEOFF.IPYNB · ANGLE 2

```
value_groups = [g["median_value_change_pct"] for _, g  
in combined.groupby("age_bucket")]  
risk_groups  = [g["risk_per_1000_90s"]          for _,  
g in combined.groupby("age_bucket")]  
h_val, p_val = stats.kruskal(*value_groups)  
h_risk, p_risk = stats.kruskal(*risk_groups)
```

IF ASKED ABOUT ALTERNATIVE="LESS"

That's a one-sided test: I expected recent-injury value change to be lower, not just different. The five age buckets have no predicted direction, so those stay two-sided, the default.

Quick facts, in case you're asked

A fast lookup, so you don't have to hold every number in your head.

11.11%	median value growth, injury-free players only, vs. 0.00% for every injured bucket
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+4.35%	no-injury median after excluding likely breakout-debut snapshots, a robustness check
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$p < 0.0001$	Mann-Whitney U, recent-injury vs. all other snapshots, statistically significant
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2.49 → 5.95	median injury risk per 1,000 90s, 21-24 bracket up to 33+, not strictly ordered
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7.14%	the only age bracket with real positive median value growth, 21-24 emerging
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582	players with both a value-growth number and a risk rate, after excluding 1,377
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22.7%	directional accuracy of the value forecast, vs. a 50% coin-flip baseline
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149.4%	median magnitude error on the same forecast
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1,959

players analyzed, same combined dataset as prior weeks

67%

of players, 1,310 of 1,959, have no matching on-field performance record

Not yet

computed: effect sizes for both tests, and the un-rounded 0.00% figure, both flagged on-slide as open items

scipy.stats

mannwhitneyu and kruskal, the tools behind both statistical tests